



Strategic forecasting of electricity demand for 100 % electrification in Malawi by 2063: A data-driven ECEEMDAN-BiGRU and quantile regression approach

Sylvester William Chisale^{a,d,*}, Han Soo Lee^{c,a,b}, Manuel Alejandro Soto Calvo^a

^a Coastal Hazards and Energy System Science Laboratory, Transdisciplinary Science and Engineering Program, Graduate School of Advanced Science and Engineering, Hiroshima University, 1-5-1 Kagamiyama, Higashi-Hiroshima, 739-8529, Hiroshima, Japan

^b Coastal Hazards and Energy System Science Laboratory, Graduate School of Innovation and Practice for Smart Society, Hiroshima University, 1-5-1 Kagamiyama, Higashi-Hiroshima, 739-8529, Hiroshima, Japan

^c Center for the Planetary Health and Innovation Science (PHIS), The IDEC Institute, Hiroshima University, 1-5-1 Kagamiyama, Higashi-Hiroshima, 739-8529, Hiroshima, Japan

^d Department of Applied Studies, Malawi University of Science and Technology (MUST), P. O. Box 5196, Limbe, Malawi

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ABSTRACT

Malawi faces significant challenges in transitioning to renewable energy due to a low electrification rate, rapid urbanization, and population growth, which drive increased electricity demand, particularly in urban transportation and residential sectors. This study uses the enhanced complete ensemble empirical mode decomposition with adaptive noise (ECEEMDAN) and bidirectional gated recurrent unit (BiGRU) models to forecast electricity demand in Malawi under three scenarios: business as usual (BAU), low demand, and high demand. These scenarios consider varying industrialization rates and policy effectiveness. By 2060, electricity demand is expected to range from 8219.47 MW in the low demand scenario to 9492.14 MW in the high demand scenario, with the BAU scenario estimating 8544.64 MW. To achieve full electrification by 2060, the BAU scenario predicts an annual increase of about 201.7 MW, while the low and high demand scenarios suggest yearly increases of 193.31 MW and 224.69 MW, respectively. These findings underscore the urgent need for strategic energy planning, infrastructure enhancement, and sustainable economic growth. The study provides valuable insights to guide policymakers and stakeholders in supporting Malawi's Agenda 2063 through effective energy development and planning.

1. Introduction

Countries that achieved full electrification primarily relied on and continue to heavily depend on fossil fuels for electricity generation. However, as global pressure to reduce CO₂ emissions intensifies, these fossil fuel-dependent nations face challenges in transitioning to renewable energy (RE) sources. African countries, despite their minimal contribution to global CO₂ emissions [1], are not exempt from the severe impacts of climate change [2] and must actively participate in the global fight against emissions. This shift in electricity generation poses significant concerns for developing nations like Malawi, where the electrification growth rate remains precarious. Compounding these challenges are the sharp increases in population and urbanization, which

contribute significantly to electricity access difficulties. In urban areas, energy demand is particularly high in the transportation sector, which is increasingly shifting towards electric vehicles and the infrastructure required for heating, cooling, and lighting [3,4].

As countries that traditionally rely on fossil fuels for electricity generation shift towards RE, low-electrification nations like Malawi must urgently increase their access rates through sustainable means. As of 2023, electric vehicles and solar PV have driven a 40 % increase in clean energy investment since 2020 [5]. Notable efforts to improve access sustainably are evident in several countries, including Kenya, Bangladesh, and India [6]. As of 2022, approximately 80 % of those without electricity reside in sub-Saharan Africa [5], with Malawi's access rate at under 18 %. Furthermore, while other African countries are

* Corresponding author. Coastal Hazards and Energy System Science Laboratory, Transdisciplinary Science and Engineering Program, Graduate School of Advanced Science and Engineering, Hiroshima University, 1-5-1 Kagamiyama, Higashi-Hiroshima, 739-8529, Hiroshima, Japan.

E-mail address: schisale@must.ac.mw (S.W. Chisale).

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making progress in improving access, it is unfortunate that, despite abundant resources, only 2 % of the world's RE investments from 2000 to 2020 were made in Africa [7], primarily due to financial challenges. However, regardless of poor access, shift in technology, and other issues including financial challenges, increase in population and urbanization is also sharply increasing electricity demand in most developing countries including Malawi.

In response to the prolonged poor electrification status in Malawi, the country has implemented various strategies to improve the situation. Notably, Malawi has developed several policies to address this issue, including the national energy policy (2018) [8], the regulatory framework for mini-grids (2019) [9], the national charcoal strategy (2017–2027) [10], the Malawi feed-in tariff policy (2012) [11], the Malawi national electrification strategy & action plan (2019) [12], and the Malawi renewable energy strategy (2017) [13], along with the introduction of independent power producers (IPPs) framework [14]. These efforts have resulted in an increase in large solar PV projects and various initiatives under the rural electrification program. Recently, new electrification targets have been established through Malawi's agenda 2063 [15], aiming for full electrification by 2063. Specifically, Malawi aims to achieve access rates of 50 %, 75 %, 80 %, 90 %, and 100 % by the years 2030, 2040, 2042, 2050, and 2060, respectively. Unfortunately, the primary energy source for cooking and heating is biomass [16]. As of 2022, the country's installed electricity capacity was approximately 546 MW [17], primarily from hydropower, followed by solar PV [18], with biomass contributing about 18.5 MW [17]. However, declining water levels have reduced capacity [19], causing outages and reducing electricity supply to meet demand [20].

Furthermore, increased efforts to improve the electricity situation in Malawi have triggered the exploration of RE resources. For instance, Chisale et al. [21,22] conducted a nationwide assessment of wind and solar PV resources and identified optimal locations for solar and wind generation. Zalengera [23] also investigated the viability of solar and wind generation on Likoma Island. Other studies have focused on the application of solar and wind energy at a small scale for community and facility use [24–26]. However, these efforts have failed to materialize due to a lack of financial backing and political will towards RE in Malawi [27,28]. Thus, to emphasize the need for diversifying the energy mix, policymakers must accurately forecast Malawi's future electricity demand to make informed decisions about these targets.

Recent advancements in energy storage technologies are pivotal for enabling reliable renewable energy integration to meet rising electricity demand [29]. Solid-state batteries enhance safety and energy density compared to conventional lithium-ion systems [30], while flow batteries and thermal storage offer scalable, long-duration grid support [32,33]. Hydrogen storage and AI-driven energy management further optimize system flexibility and resilience [31,34]. Green hydrogen is projected to account for 3 % of global final energy use by 2030, rising to 12 % by 2050 [35], making it a cornerstone of decarbonization strategies. Smart grids and demand-side management reduce peak loads and improve load balancing [36,37]. Despite progress, material scarcity and battery recycling pose environmental and economic challenges. Ongoing research, investment, and policy support are critical for advancing sustainable and resilient energy systems.

Several electricity forecasting techniques, including advanced data-driven methods like artificial neural networks (ANN) [38] and deep learning [39], as well as simpler approaches such as linear regression [40], time series analysis [41,42], grey models [43,44], and system dynamics [45], can effectively predict electricity demand with varying characteristics. A growing number of studies have demonstrated that combining multiple approaches yields good results. For instance, Olu-Ajayi et al. [46] integrated gradient boosting (GB), deep neural networks (DNN), random forest (RF), K-nearest neighbors (KNN), support vector machines (SVM), decision trees (DT), and linear regression (LR). Peng et al. [47] applied enhanced wavelet transform (EWT)-attention long short-term memory (LSTM) networks to enhance prediction

accuracy. Additionally, Zhussupbekov et al. [48] combined multiple regression (MR), support vector machines (SVM), and ANN, while Gao et al. [49] utilized KNN, decision trees (DT), XGBoost, gaussian process regression (GPR), and ANN. Dasi et al. [50] focused on optimized CatBoost, incorporating ANN methodologies to improve forecasting performance.

The integration of forecasting methods with decomposition techniques is increasingly popular, demonstrating significant advancements in accuracy and efficiency across various applications, including energy-related studies [51–56]. For instance, Ding [57] combined ensemble empirical mode decomposition (EEMD) with ANN. Ghimire et al. [58] developed a framework using ANN, encoder-decoder long-term memory (EDLSTM), and improved complete ensemble empirical mode decomposition (ICMD). Oqaibi & Bedi [59] introduced a method that combines gaussian smoothing, CEEMDAN, and an attention-based bidirectional gated recurrent unit (BiGRU) model. Chreng et al. [60] created a hybrid model for electricity demand forecasting using improved EEMD and recurrent neural networks (RNN) with ERA5 climate variables.

Integrating Enhanced Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (ECEEMDAN) and Bidirectional Gated Recurrent Units (BiGRU) offers a robust framework for electricity demand forecasting. ECEEMDAN decomposes time series data into intrinsic mode functions (IMFs), isolating trends and oscillatory components to reduce noise and simplify the dataset for modeling [61,62]. BiGRU complements this by analyzing temporal dependencies in both forward and backward directions, capturing complex patterns and non-linear relationships. This dual perspective enables the model to extract meaningful features and discern critical temporal patterns, significantly boosting forecasting accuracy. Together, these methods offer a reliable and advanced tool for addressing the complexities of dynamic and volatile datasets like electricity demand, outperforming traditional forecasting approaches.

ECEEMDAN-LSTM approaches have been successfully applied for electricity forecasting in Cuba [62]. Limited studies have been conducted in Malawi to comprehensively assess electricity demand in line with Malawi's Agenda 2063. To the authors' knowledge, only two studies have attempted to forecast Malawi's demand up to 2030: Chaima et al. [63], who modeled electricity demand up to 2030, and Tauro et al. [64], who used the model for analysis of energy demand (MAED), both demonstrating increasing trends. Furthermore, despite the rise of advanced forecasting methods, there may be few studies employing machine learning techniques for electricity demand forecasting in Malawi.

Addressing these challenges requires strategic forecasting of electricity demand, which is essential for Malawi to achieve its goal of 100 % electrification by 2063. Accurate predictions will enable policymakers and stakeholders to make informed decisions regarding infrastructure development, investment in RE technologies, and the efficient allocation of resources. This study utilized a hybrid approach for long-term electricity demand forecasting by integrating ECEEMDAN, BiGRU networks and quantile regression technique. This hybrid approach effectively captures the complexity and nonlinearity of demand data, offering significant improvements over traditional models in handling forecasting uncertainties [62,65]. By leveraging advanced modeling techniques and historical data, the research aims to provide a robust framework for understanding future energy needs within Malawi's socio-economic context. The findings of this research are expected to significantly contribute to national electrification strategies and policy formulation, paving the way for a sustainable and resilient energy future. The main contributions of this paper are described as follows.

- i. *Development of a hybrid ECEEMDAN-BiGRU framework for time series forecasting, leveraging the strengths of empirical mode decomposition to handle non-stationarity and BiGRU's ability to capture complex temporal dependencies and bidirectional patterns in demand signals.*

- ii. *Integration of quantile regression with deep learning outputs* to construct probabilistic forecasts, enabling the derivation of uncertainty bounds that provide a more robust representation of predictive confidence intervals.
- iii. *Design of a long-horizon forecasting pipeline* that incorporates pre-processed socio-economic variables (e.g., interpolated population and GDP projections) into the deep learning model as exogenous regressors, enhancing its ability to capture long-term trends.
- iv. *Demonstration of the decomposition-reconstruction approach* using Intrinsic Mode Functions (IMFs) derived from ECEEMDAN, improving signal clarity and forecast accuracy compared to raw time series modeling.
- v. *Development of a data-driven forecasting framework tailored to energy transition contexts*, offering strategic insights to support long-term energy planning and policy formulation in developing countries such as Malawi.

This study offers valuable insights into accelerating Malawi's electrification efforts and promoting sustainable energy solutions, aligning with broader aspirations for a greener future. The paper is divided into five sections: Section 2 outlines the materials and methods, Section 3 presents the research findings, Section 4 provides a general discussion, and Section 5 concludes the study.

2. Materials and methods

2.1. Study region

Malawi, a landlocked country in southeastern Africa, spans approximately 118,484 square kilometers. Its diverse topography ranges from lowland plains to high plateaus and mountains. The climate is sub-tropical, marked by a rainy season from November to April and a dry season from May to October. Average temperatures vary widely depending on the altitude, ranging from 20 °C to 27 °C, while peak temperatures can reach up to 35 °C during the hottest months. Dominating its geography is Lake Malawi, one of Africa's Great Lakes, which is central to the local ecosystem and economy, supporting vital fisheries and supplying the bulk of the country's hydroelectric power.

The electricity sector is overseen by the Electricity Supply Commission of Malawi (ESCOM) and the Electricity Generation Company

(EGENCO), which manage electricity distribution and power generation, respectively. The Malawi Energy Regulatory Authority (MERA) regulates this sector, setting tariffs and ensuring fair practices. Additionally, the introduction of feed in tariff [11] has broadened the energy mix with solar projects, improving energy security and reducing reliance on hydro sources, which are vulnerable to climate-related fluctuations.

With limited data available, records from January 2022, taken every 30 min, were analyzed to establish Malawi's average daily load profile. This resulted in 1488 data points, as illustrated in Fig. 1. Electricity consumption begins increasing at 6 a.m., continuing prominently throughout daylight hours with notable peaks around lunch and dinner times. At 11:00 a.m., the average demand reaches 244.54 MW, peaking at 331.99 MW, primarily driven by heightened household activities such as cooking and appliance use. The evening peak at 19:00 is particularly significant, where the average demand escalates to 253.88 MW and spikes to a maximum of 342.18 MW, predominantly due to residential lighting and dinner preparation activities. These dinner-time peaks underscore the substantial impact of household energy use, accounting for approximately 42 % of total demand. During this critical period, the extensive use of both lighting, entertainment and cooking appliances highlights the combined effect of residential energy consumption. Post-dinner, there is a noticeable decline in demand as cooking activities cease and the use of lighting and entertainment appliances gradually decreases while households are sleeping. Additionally, the minimum values in the load profile highlight instances of power outages within the system.

2.2. Data and preprocessing

The study utilized electricity generation data provided by ESCOM, available only monthly due to data limitations, rather than the preferred daily or hourly format. This dataset spans the national monthly peak generation from 2000 to 2022 with 276 monthly data points. In 2019, only 71.8 % of Malawi's total electricity demand was met [15], largely because the country's hydropower-dependent generation often faces significant reductions due to low water levels or infrastructural damage from flooding. To address data irregularities caused by these fluctuations, outliers in the dataset were replaced with annual averages, ensuring a more consistent analysis.

The data sources for this study cover a range of variables essential for

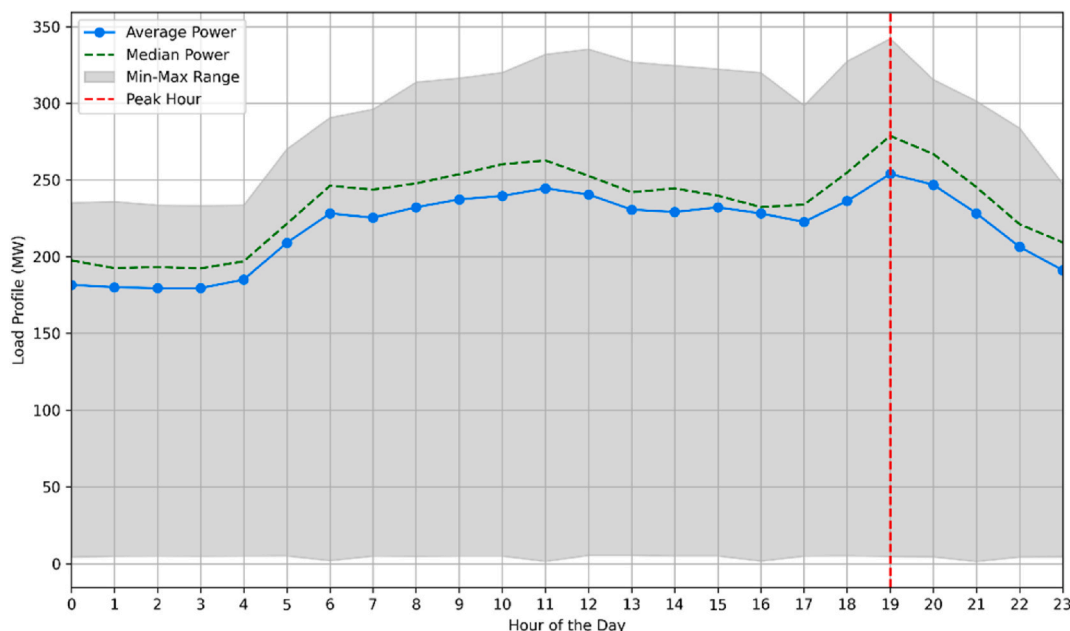


Fig. 1. Average Daily Load Profile for Malawi: Hourly averages, medians, minimums, and maximums of electricity demand (January 2022).

analyzing electricity demand and environmental conditions in Malawi. Annual data on access rates, spanning from 2000 to 2022, was obtained from Our World in Data [66], while population projections up to 2100 and historical gross domestic product (GDP) data from 2000 to 2022 were sourced from Macrotrends [67,68]. Monthly electricity generation data, crucial for demand assessment, was provided by ESCOM from 2000 to 2022. Additionally, climate variables, including temperature and relative humidity, both on a 0.25-degree ERA5 scale, were sourced from the World Bank [69], with temperature data available from 1950 and relative humidity from 2000, offering valuable insights into long-term environmental trends. The input variables were compared, as shown in Figures A1–2.

2.3. Normalization of the input data

This study utilizes predictors such as climatic variables (temperature and relative humidity), population, GDP, and the percentage electricity access rate. All these predictors, including the power demand, were normalized before being input into the simulation model to enhance the training accuracy of the neural network. All input and target values were scaled to a range between 0 and 1, as outlined in Eq. (1). Thus, X_{norm} is a normalized value in a customized range of [a, b] for the original data point X , where X_{min} and X_{max} are the minimum and maximum values in the original dataset, respectively.

$$X_{norm} = a + \left(\frac{X - X_{min}}{X_{max} - X_{min}} \right) \times (b - a) \quad (1)$$

2.4. Decomposition techniques (ECEEMDAN)

Enhanced Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (ECEEMDAN) is an advanced signal decomposition method that enhances the capabilities of previous EMD variations through adaptive noise handling and iterative filtering [61]. The method consists of three key components [61].

- i. **Window Size Determination:** The calculation of the window size in the ECEEMDAN algorithm is based on the statistical properties of the signal, such as autocorrelation, spectral entropy, and noise level. The window size is crucial for calculating the moving standard deviation and for signal smoothing, which helps in accurately identifying the IMFs.
- a. **Estimation of the Dominant Period:** First, the dominant period of the signal $x(t)$ is estimated using the periodogram method. The periodogram estimates the power spectral density (PSD) of the signal $x(t)$ as in Eq. (2). Here, N is the number of samples in the signal and f is the frequency.

$$PSD = P(f) = \frac{1}{N} \left| \sum_{t=0}^{N-1} x(t) e^{-i2\pi ft} \right|^2 \quad (2)$$

The index of the maximum spectral power is found as in Eq. (3).

$$idx = \arg \max_{f>0} P(f) \quad (3)$$

The dominant period T_d is then calculated as the inverse of the frequency corresponding to the maximum power as in Eq. (4).

$$T_d = \frac{1}{f(idx)} \quad (4)$$

If the frequency corresponding to the maximum power is zero or no significant peak is found, the total length of the signal is used as the dominant period.

- b. **Estimation of the Noise Level:** The noise level of the signal is estimated using the median absolute deviation (MAD) of the

successive differences of the signal as in Eq. (5), where $\Delta x = x(t+1) - x(t)$.

$$MAD = \text{median}(|\Delta x - \text{median}(\Delta x)|) \quad (5)$$

Assuming Gaussian noise (normally distributed), the normalized noise level σ_n is computed as in Eq. (6), where $\sigma_x = \text{std}(x)$ is the standard deviation of the signal x .

$$\sigma_n = \frac{MAD}{0.6745 \cdot \sigma_x} \quad (6)$$

- c. **Calculation of the First Zero-Crossing of the Autocorrelation Function:** The autocorrelation function $R(\tau)$ of the signal $x(t)$ is used to calculate the first zero-crossing as in Eq. (7).

$$R(\tau) = \frac{1}{N} \sum_{t=1}^{N-\tau} x(t)x(t+\tau) \quad (7)$$

The first zero-crossing is found as in Eq. (8).

$$Z_0 = \min(\tau : R(\tau) < 0) \quad (8)$$

If no zero-crossing is found, the total length of the signal is used.

- d. **Calculation of the Window Size:** Finally, the window size W is computed by considering the intrinsic properties of the signal to accurately reflect and capture local and global phenomena. The computation of W can be then expressed as in Eq. (9), where ε is a small constant to avoid logarithm getting undefined.

$$W = \lceil \log(T_d + \varepsilon) \cdot (1 + \sigma_n) \cdot Z_0 \rceil \quad (9)$$

where, T_d is the dominant period, σ_n is the normalized estimated noise level, and Z_0 is the first zero-crossing of the autocorrelation function.

The window size is dynamically adjusted to optimize the decomposition's accuracy, tailoring itself to the complexity of the signal. This approach enhances the detection of modes.

ii. Adaptive Noise Generation

The noise intensity $Nstd$ is computed using several signal properties, including the moving standard deviation, spectral entropy, and a signal-to-noise ratio (SNR) estimate.

a. Moving Standard Deviation Calculation

The moving standard deviation $\sigma_{mov}(x)$ of the signal $x(t)$ over a specified window size W is calculated as in Eq. (10). Here, μW is the moving average over the windows size W .

$$\sigma_{mov}(x) = \sqrt{\frac{1}{W} \sum_{i=0}^{W-1} (x(t+i) - \mu W)^2} \quad (10)$$

b. Spectral Entropy Calculation

The spectral entropy H_s of the signal is calculated from its power spectral density (PSD) $P(f)$ (described in Eq. (1)) using the Welch method as in Eq. (11), where ε is a small constant to avoid logarithm getting undefined:

$$H_s = - \sum \frac{P(f)}{\sum P(f)} \log_2 \left(\frac{P(f)}{\sum P(f)} + \varepsilon \right) \quad (11)$$

c. Signal-to-Noise Ratio (SNR) Estimate

The SNR is estimated based on the variance of the signal $x(t)$ and the variance of its differences as in Eq. (12), where $\Delta x = x(t+1) - x(t)$.

$$SNR = 10 \log_{10} \left(\frac{\text{Var}(x)}{\text{Var}(\Delta x)} \right) \quad (12)$$

d. Scaling Factor Calculation

The noise intensity scaling factor α is computed as in Eq. (13), where N is the number of samples in the signal. This factor balances the contribution of entropy and SNR to the noise intensity.

$$\alpha = \left(1 - \frac{H_s}{\log_2 N} \right) (1 + e^{-SNR/10}) \quad (13)$$

e. Base Noise Intensity and Adaptive Scaling

The base noise intensity I_{base} is derived from the mean moving standard deviation and computed as in Eq. (14).

$$I_{base} = 1 - \text{mean}(\sigma_{mov}(x)) \quad (14)$$

The final noise intensity N_{std} is calculated by applying the adaptive scaling factor α as in Eq. (15).

$$N_{std} = I_{base} \cdot \alpha \quad (15)$$

To ensure the noise intensity remains within a reasonable range, it is constrained to $0.01 < N_{std} < 0.5$. A too small intensity might lead to mode mixing, while a too large noise intensity will result in altering the original signal in a way that loses its properties.

iii. Generation of the Original Signal with White Noise

White noise is added to the original signal to create an ensemble for decomposition. The white noise $n(t)$ is generated with a standard deviation proportional to N_{std} as in Eq. (16), where, $\mathcal{N}(0, 1)$ represents a normal distribution with zero mean and unit variance.

$$n(t) = N_{std} \cdot \sigma_{mov}(x) \cdot \mathcal{N}(0, 1) \quad (16)$$

A noisy signal $x_i(t)$ is created for each ensemble realization by adding the white noise $n(t)$ to the original signal $x(t)$, as expressed in Eq. (17).

$$x^{(i)}(t) = x(t) + n(t) \quad (17)$$

This noisy signal $x_i(t)$ is used in each realization of the ECEEMDAN algorithm to perform the decomposition iteratively, thereby reducing mode mixing and improving the quality of extracted IMFs.

iv. Mode Extraction Process

Once the noisy signal is generated by adding white noise to the original signal, the next step in the ECEEMDAN algorithm involves calculating the first mode from each noisy realization. This process aims to extract the Intrinsic Mode Functions (IMFs) iteratively by averaging the modes obtained from multiple noisy versions of the signal. For $i = 1 : NR$ where NR is the number of realizations, $x_i(t)$ (the original signal with added noise), the first mode $IMF_{1,i}$ is computed using the Empirical Mode Decomposition (EMD) method. After calculating $IMF_{1,i}$ for each noisy realization, the modes are averaged to estimate the first ‘true’ mode as in Eq. (18).

$$IMF_1 = \frac{1}{NR} \sum_{i=1}^{NR} IMF_{1,i} \quad (18)$$

Once the first mode IMF_1 is calculated, it is subtracted from the original signal $x(t)$ to produce the first residue $r(t)$ as in Eq. (19).

$$r(t) = x(t) - IMF_1 \quad (19)$$

This residue $r(t)$ now represents the original signal with the first mode

removed. After each IMF is extracted, the residual signal is filtered to eliminate the residual noise before adding noise again.

The residue will be used as the new signal to generate the next set of noisy realizations in subsequent iterations. This process is repeated for $j = 1 : \max(\text{iter})$ until all significant modes are extracted, and the residue becomes a monotonic function, or a stopping criterion is met. The stopping criteria is based on maximum absolute difference between the current and the previous mode or consecutive iteration without convergency. This enhanced approach significantly reduces mode mixing issues and computational time while maintaining decomposition accuracy, making it particularly suitable for analysis of complex signals in power systems and other engineering applications.

2.5. Bidirectional gated recurrent unit (BiGRU)

The preprocessed data, decomposed into IMFs using ECEEMDAN, was split into 80 % for training and 20 % for testing. The bidirectional gated recurrent unit (BiGRU) enhances traditional GRUs [70] by processing input sequences in both forward and backward directions, improving the model’s ability to capture complex temporal dependencies. BiGRUs utilize two parallel GRU layers: one processes the sequence from past to future, and the other from future to past. This dual-directional approach allows the model to capture patterns influenced by both past and future data, which is crucial for tasks such as time-series prediction and natural language processing.

GRUs use two key gates to regulate information flow: the update gate and the reset gate. The update gate (z_t) (Eq. (20)) controls how much of the previous hidden state is carried forward or updated. It is calculated using the sigmoid function (σ), with weights (W_z), input vector (x_t), the previous hidden state (h_{t-1}), and bias (b_z). The reset gate (r_t) (Eq. (21)) manages how much past information to forget, also using the sigmoid function, with its own weights (W_r) and bias (b_r). These gates allow the GRU to handle both short-term and long-term dependencies effectively.

In BiGRU, these mechanisms operate in both directions. The candidate activation ($\tilde{h}_t$) (Eq. (22)) combines the reset gate’s filtered previous hidden state with the current input using a hyperbolic tangent function ($\tanh$). The final hidden state (h_t) (Eq. (23)) is a blend of the old state and the candidate state, moderated by the update gate, allowing the GRU to efficiently manage information across multiple time steps.

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z) \quad (20)$$

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r) \quad (21)$$

$$\tilde{h}_t = \tanh(r_t * W_a[h_{t-1}, x_t] + b_a) \quad (22)$$

$$h_t = (1 - z_t) * \tilde{h}_t + z_t * h_{t-1} \quad (23)$$

2.6. Development of the ECEEMDAM-BiGRU model

The proposed approach involves decomposing the original demand data into multiple intrinsic modes using the ECEEMDAN method. These modes are then used as input for an attention-based BiGRU model to perform the prediction task. A detailed step-by-step explanation of the method is provided below.

- i. Data Decomposition: The original demand data is decomposed into multiple IMFs and a residual component using the ECEEMDAN technique. This step simplifies the time-series data, isolating different modes of variation for more focused predictions.
- ii. Normalization and Input Structuring: Next, the input features (e.g., population, GDP, temperature) and target values (IMFs) are normalized to ensure consistency. The normalized data is then structured into sequences with three dimensions: number of timestamps (S), sequence length (W), and features per sequence (F) for input into the BiGRU model.

- iii. Performance Validation: The data is split into training, validation, and testing sets to assess model performance. The training and validation sets are used to build and fine-tune the model, while the test set evaluates its generalization accuracy.
- iv. Training Attention-Based BiGRU Models: Attention-based BiGRU models are trained on each IMF and residual component. Bidirectional GRU layers capture dependencies in both forward and backward directions, and an attention mechanism highlights relevant time steps. Hyperparameter tuning is applied to optimize the model.
- v. Prediction and Aggregation: The trained BiGRU models are used to predict on the test data, and the final forecast is obtained by aggregating the individual predictions from each IMF and residual.

2.7. Statistical evaluation

The performance of the models was evaluated using several statistical metrics, including Mean Squared Error (MSE), Mean Absolute Error (MAE), R-squared (R^2), Mean Absolute Percentage Error (MAPE), and Bias. Eqs. (24)–(28) were used to calculate these metrics for all n variables in both the model (M) and the historical (O) whereas $\bar{O}_i$ and $\bar{M}_i$ are their respective means.

$$MSE = \frac{1}{n} \sum_{i=1}^n (M_i - O_i)^2 \quad (24)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |M_i - O_i| \quad (25)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (M_i - O_i)^2}{\sum_{i=1}^n (O_i - \bar{O}_i)^2} \quad (26)$$

$$MAPE = \frac{100}{n} \sum_{i=1}^n \left| \frac{M_i - O_i}{O_i} \right| \quad (27)$$

$$Bias = \frac{1}{n} \sum_{i=1}^n (M_i - O_i) \quad (28)$$

2.8. Future demand analysis

This study forecasted future electricity demand in Malawi, aligning with the government's Agenda2063 electrification targets set for the years 2030 (50 % electrification), 2040 (75 %), 2042 (80 %), 2050 (90 %), and 2060 (100 %). Outputs from the ECEEMDAN-BiGRU model were used to estimate the annual average electricity demand for each milestone, considering projected access rates. The analysis incorporated future population projections to calculate per capita electricity demands, which were then aggregated to determine the total national demand. It was assumed that initially, generated power would extend access to only 12.4 % of the population, reflecting the baseline access level in 2020.

2.9. Future demand scenarios using quantile regression

The study implemented quantile regression to develop three distinct scenarios: business-as-usual, low demand, and high demand scenarios. Quantile regression models the relationship between covariates and specific conditional quantiles of an outcome, providing a robust estimate of its distribution. Quantiles indicate the share of data above or below defined thresholds [71].

This approach uses quantile regression to model a robust forecast range for future electricity demand, capturing both central trends and

variability across different quantile levels. Unlike traditional mean-based models, quantile regression estimates conditional quantiles for demand at various levels, offering a refined understanding of potential fluctuations. For this study, the lower bound is represented by the 5th percentile ($q = 0.05$), identifying the demand level below which only 5 % of observations are anticipated, while the upper bound is set by the 99.9th percentile ($q = 0.999$) indicating that 99.9 % of observations should fall below this threshold. In quantile regression, the goal is to estimate conditional quantiles $Q_q(Y|X)$ for demand Y based on a predictor X by minimizing an asymmetric loss function using Eq. (29) [72, 73].

$$Q_q(Y|X) = \min_{\alpha, \beta} \sum_{i=1}^n \rho_q(Y_i - \alpha - \beta X_i) \quad (29)$$

where Y_i and X_i are the actual and predicted demand values, respectively, and the quantile check function $\rho_q(u)$ is defined using Eq. (30).

$$\rho_q(u) = \begin{cases} qu, & u \geq 0 \\ (q-1)u, & u < 0 \end{cases} \quad (30)$$

This loss function asymmetrically penalizes errors depending on the quantile level q , allowing the model to capture demand variations across quantiles. The resulting quantile functions are defined using Eqs. (31) and (32).

$$Q_{0.05}(Y|X) = \alpha_{0.05} + \beta_{0.05}X \quad (31)$$

$$Q_{0.999}(Y|X) = \alpha_{0.999} + \beta_{0.999}X \quad (32)$$

where α and β are the intercept and slope for each quantile. These expressions create a flexible range around each forecasted demand point, reflecting the expected variability in electricity demand. The constructed interval between the 5th and 99.9th percentiles provides a comprehensive forecast range, offering a clear visual and quantitative representation of potential demand variability. By establishing an expected range of demand values, this method supports more resilient, well-informed decision-making for energy system planning and forecasting, equipping stakeholders to manage uncertainty in demand projections effectively. The overall study approach utilizing different models is illustrated in Fig. 2.

3. Results and discussions

3.1. Selected input variables

The study expanded its analytical scope by incorporating demographic and climatic variables to enhance the accuracy of its predictions. A statistical analysis using linear regression was conducted to explore the relationships between temperature, relative humidity, population, electricity access rate, and GDP with respect to electricity demand. The correlation results for all input variables are depicted in Fig. 3. The analysis revealed that climate-related variables do not have a significant direct influence on electricity demand. Specifically, the correlation coefficients for temperature and relative humidity with respect to electricity demand were found to be 5.1 % and –17 %, respectively, indicating minimal influence. In contrast, demographic variables such as population, GDP, and electricity access rate demonstrated strong correlations with electricity demand. Specifically, the population showed a correlation of 92 %, GDP 91 %, and electricity access rate 81 %, underscoring their substantial influence on the demand dynamics. These findings highlight the critical role of socioeconomic factors in shaping electricity demand patterns, guiding targeted strategies for energy policy and infrastructure development.

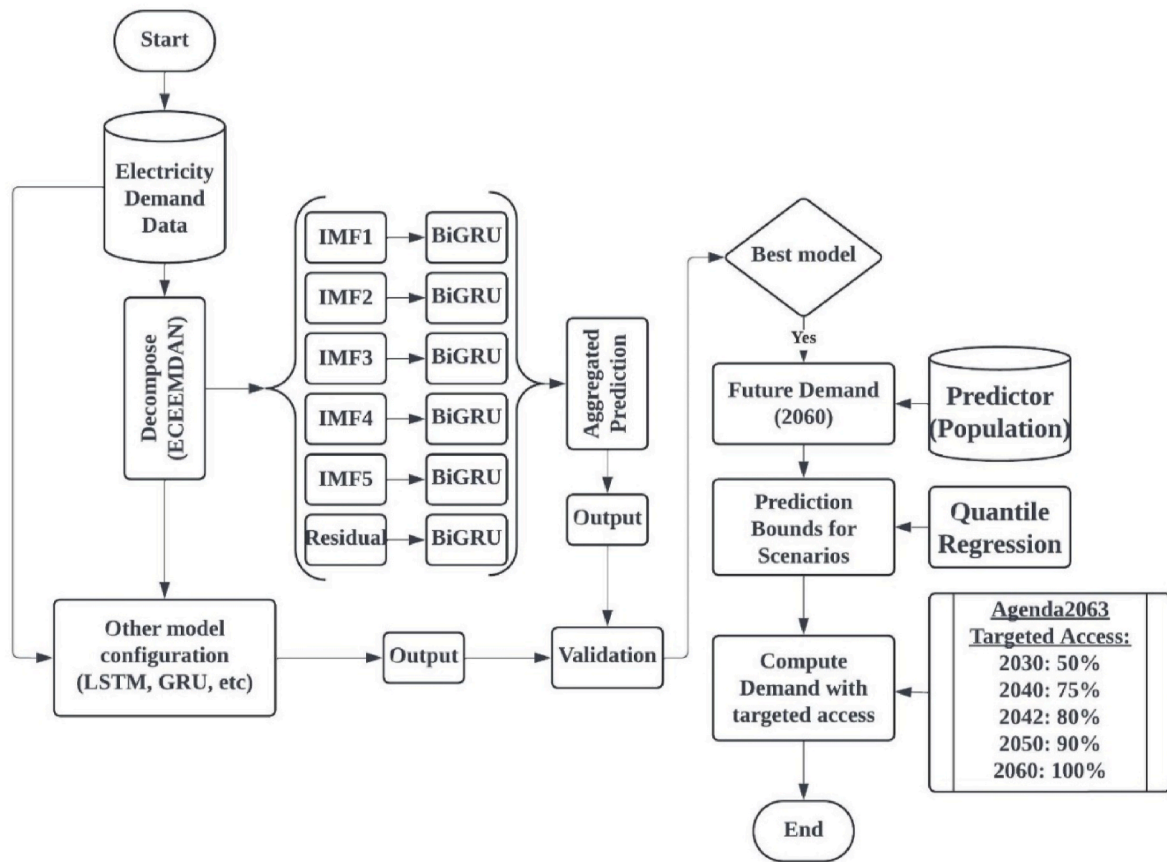


Fig. 2. Overall study approach utilizing the combined ECEEMDAN and BiGRU with quantile regression for power demand prediction.

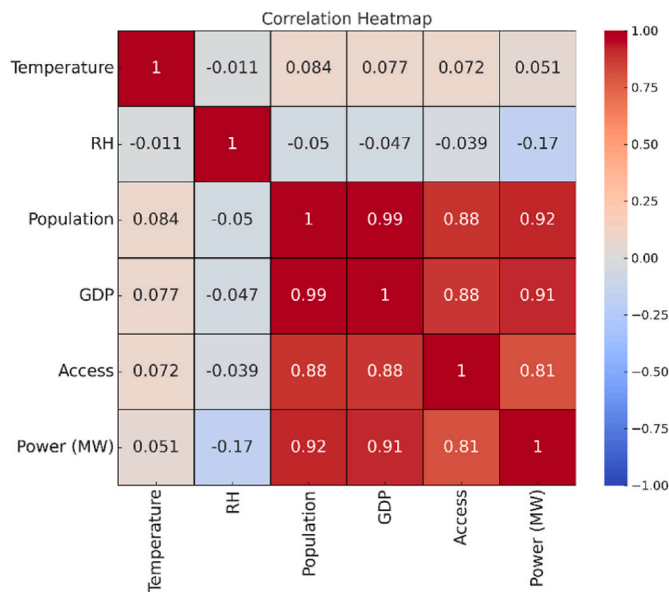


Fig. 3. Comparison of input variable data (generation, access rate, GDP, population, relative humidity, and temperature) using correlation metrics.

3.2. Decomposed electricity demand

The decomposition of electricity demand using the ECEEMDAN yielded five IMFs and a residual signal, as illustrated in Fig. 4. Given that the data analyzed were monthly, IMF3 captured the annual or seasonal variations in electricity demand. The residual signal offered insights into

the long-term trend of electricity demand across the study period, highlighting a significant increase over the years. The IMFs underwent further statistical analysis using a log-log plot to assess the energy density and periods of the IMFs. This analysis ensures optimal separation by emphasizing proper period spacing and spectral distinction between Intrinsic Mode Functions (IMFs). The method underscores that IMFs should exhibit sufficient energy density and maintain a clear period separation, with at least one period distance between modes, to effectively capture distinct trends [61]. These IMFs, along with residual signals, reveal critical patterns in both short-term and long-term electricity demand.

3.3. Prediction experiments and model choice

The determination of the optimal configuration for the electricity demand prediction model necessitated the evaluation of various setups, which included implementations employing LSTM, GRU, ECEEMDAN-LSTM, ECEEMDAN-GRU, and ECEEMDAN-BiGRU architectures, both with and without the inclusion of climatic variables (temperature, relative humidity) and demographic variables (population, GDP, access). The 'ReLU' activation function consistently yielded the best results across all configurations, and the 'Adam' optimizer with a learning rate of 1E-06 was identified as the most effective choice. This setup was consistently optimal across models, supporting enhanced predictive accuracy and model stability in forecasting electricity demand. Detailed performances of these models are outlined in Table 1, which presents a comparative analysis, while a Taylor diagram (Fig. 6) was utilized to select optimal model (used normalized correlation coefficients ranging from 0 to 1 for clarity).

The projected electricity demand presents a significant challenge for Malawi. Based on historical data, the estimated demand for 2022 is

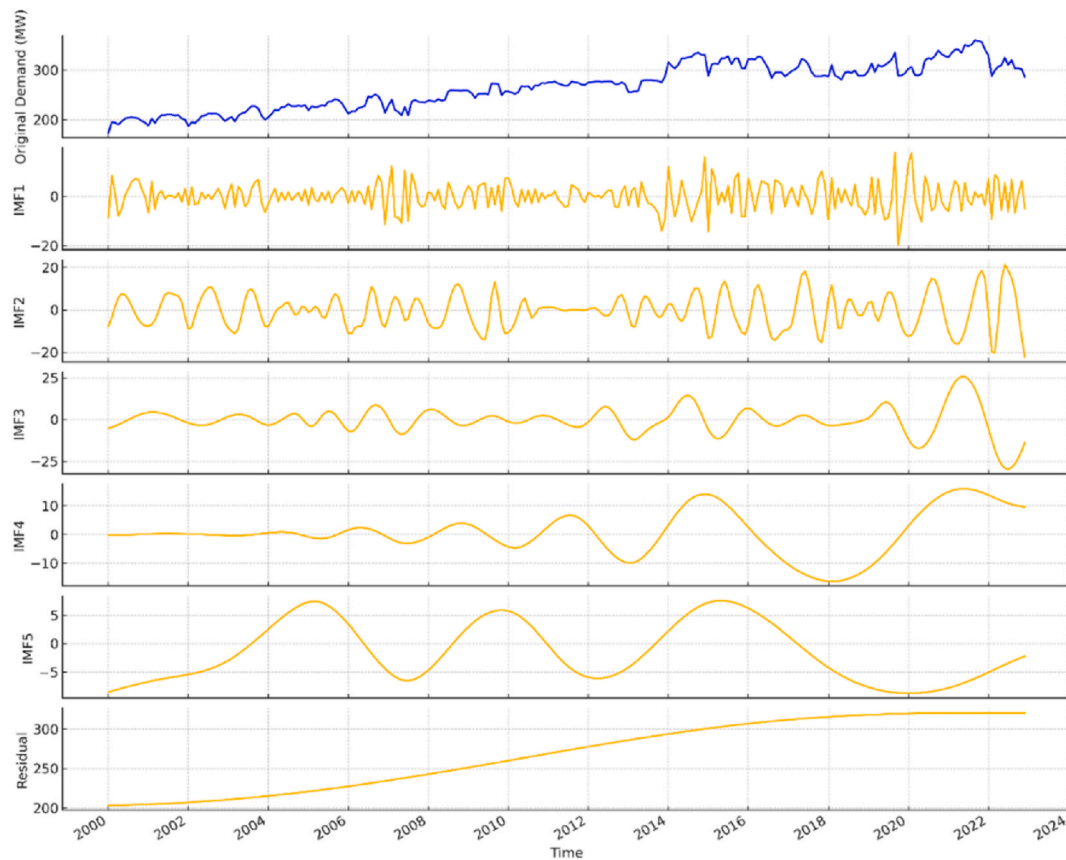


Fig. 4. Decomposition of electricity demand using ECEEMDAN into 5 Intrinsic Mode Functions (IMFs) and a residual signal.

Table 1

Statistical performances of different model combinations based on ECEEMDAN decomposition and RNN models (Deep blue = best, deep red = worst).

Model	Climate (Temp, RH)	MSE	MAE	R ²	MAPE	Bias	ΔSTD
LSTM	Yes	67.07	6.16	0.9636	0.0224	-2.79	1.242
LSTM	No	96.04	7.66	0.9478	0.0276	-5.96	2.292
GRU	Yes	67.07	6.156	0.9636	0.0224	-2.79	1.242
GRU	No	93.42	7.65	0.9492	0.028	-5.01	0.162
ECEEMDAN-GRU	Yes	76.23	6.56	0.9586	0.0243	-0.59	1.562
ECEEMDAN-GRU	No	97.97	7.91	0.9468	0.0289	-4.59	0.465
ECEEMDAN-LSTM	Yes	83.15	6.75	0.9548	0.0247	-1.5	1.782
ECEEMDAN-LSTM	No	66.38	5.89	0.9639	0.0214	-1.25	1.742
ECEEMDAN-BiGRU	Yes	52.45	5.53	0.9715	0.0205	-0.68	1.033

approximately 434.4 MW. With ongoing efforts towards electrification and population growth, the required demand is projected to reach 570.9 MW by 2030, 779.3 MW by 2040, 794.5 MW by 2042, 934.5 MW by 2050, and 1059.5 MW by 2060. Despite these projections, this demand currently serves about 12.4 % of the population, highlighting a considerable gap in electricity access. The data underscores a clear upward trend in demand, emphasizing the urgent need for strategic expansions in energy production capacity to meet future needs and support broader access to electricity across the country.

Among the diverse models assessed, the ECEEMDAN-BiGRU model distinguished itself by demonstrating superior accuracy in key statistical measures. Integrating ECEEMDAN model with BiGRU, this model recorded the lowest MSE of 52.45 and the highest R² value of 0.9715,

signaling exceptional prediction precision. Moreover, it achieved a notably low MAE of 5.53 and MAPE of 0.0205, surpassing other models in terms of forecast accuracy. With a minimal bias of -0.68, this model reliably captures the variability in electricity demand. By effectively leveraging both climatic and demographic inputs, the BiGRU model serves as a robust tool for strategic electricity demand forecasting.

3.4. Prediction with ECEEMDAN-BiGRU model

This study employed BiGRU modeling and quantile regression to forecast future electricity demand and quantify uncertainty. The BiGRU model was optimized through hyperparameter tuning and trained on time series data with predictors like population, GDP, temperature, and

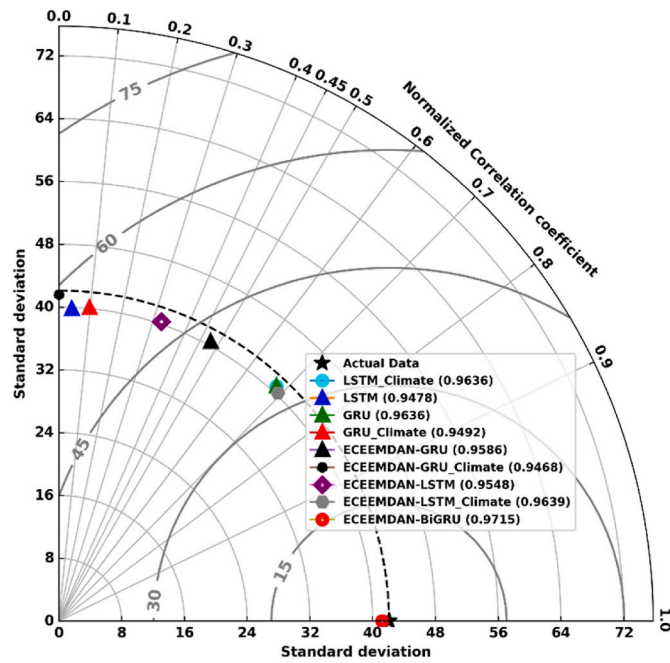


Fig. 6. Taylor diagram for model selection, showing the normalized correlation coefficient (actual correlation coefficients are provided in the legend).

historical demand. The model achieved a high degree of accuracy, reflected by MSE of 52.45, MAE of 5.53, and an R^2 value of 0.9715. These metrics demonstrate the model's reliability in capturing demand patterns. The MAPE was 0.0205, and the Bias was -0.68 , indicating minimal underestimation. Additionally, the standard deviation of predicted values (41) closely aligned with the actual values (42.1), underscoring the model's robustness.

Malawi, like most African countries, is experiencing a sharp increase in electricity demand [74,75]. The projected demand poses a significant challenge. Based on historical data, the estimated demand for 2022 is approximately 434.4 MW. With ongoing efforts toward electrification

and population growth, the required demand is projected to reach 570.9 MW by 2030, 779.3 MW by 2040, 794.5 MW by 2042, 934.5 MW by 2050, and 1059.5 MW by 2060. Despite these projections, this demand would still serve about 12.4 % of the population, highlighting a substantial gap in electricity access. The data underscores a clear upward trend in demand, emphasizing the urgent need for strategic expansions in energy production capacity to meet future needs and support broader electricity access across the country.

To further enhance the forecast's reliability, quantile regression was applied to establish a range of potential demand fluctuations, with the 5th and 99.9th percentiles serving as lower and upper bounds, respectively. This quantile-based interval offers a clear view of variability in demand, capturing both typical and extreme scenarios. By integrating BiGRU forecasting with quantile-based uncertainty intervals, the study delivers robust demand projections that account for future uncertainties, aiding in data-driven energy planning. The forecasted demand up to 2060, quantile-based demand bounds, and model performance against historical data are depicted in Fig. 7.

3.5. Demand with respect to proposed access rates

In line with Malawi's Agenda 2063, an in-depth analysis was conducted to project electricity demands under various scenarios for critical target years as the nation advances toward higher electrification rates. The scenarios evaluated include a business as usual (BAU) Scenario representing standard demand projections, a low demand scenario depicting lower possible outcomes, and a high demand scenario illustrating potential upper limits of electricity demand. Fig. 8 illustrates the projected electricity demand in relation to specific electrification targets and compares these projections with other potential electrification scenarios. This comparison helps to visualize the range of possible outcomes, highlighting how different levels of national electrification could impact overall energy demand in the context of Malawi's Agenda 2063. The scenarios examined include.

- **2030 Projections:** For the year 2030, with a 50 % electrification rate, the BAU scenario projects a demand of 2435.27 MW. The low demand scenario suggests a minimum demand of 2363.17 MW,

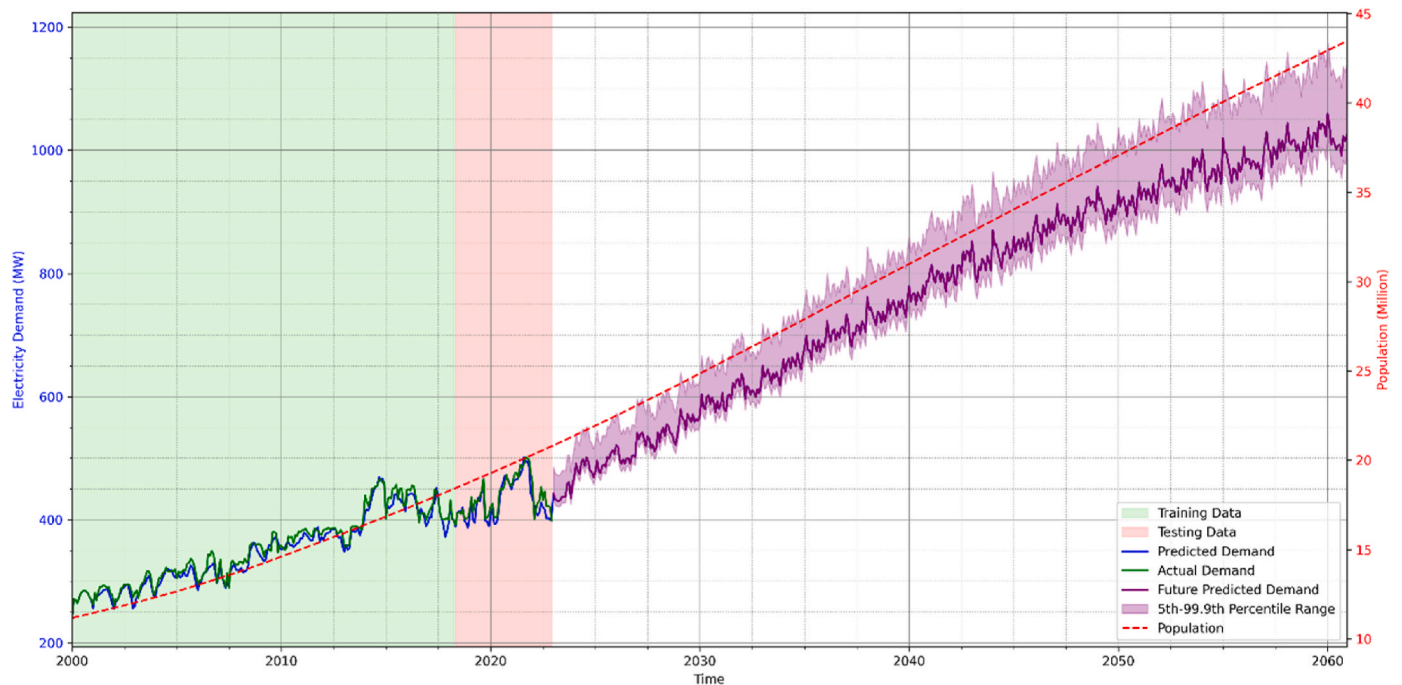


Fig. 7. Demand and population trends, showing historical data and future predictions with confidence intervals on future demand, for the period from 2000 to 2060.

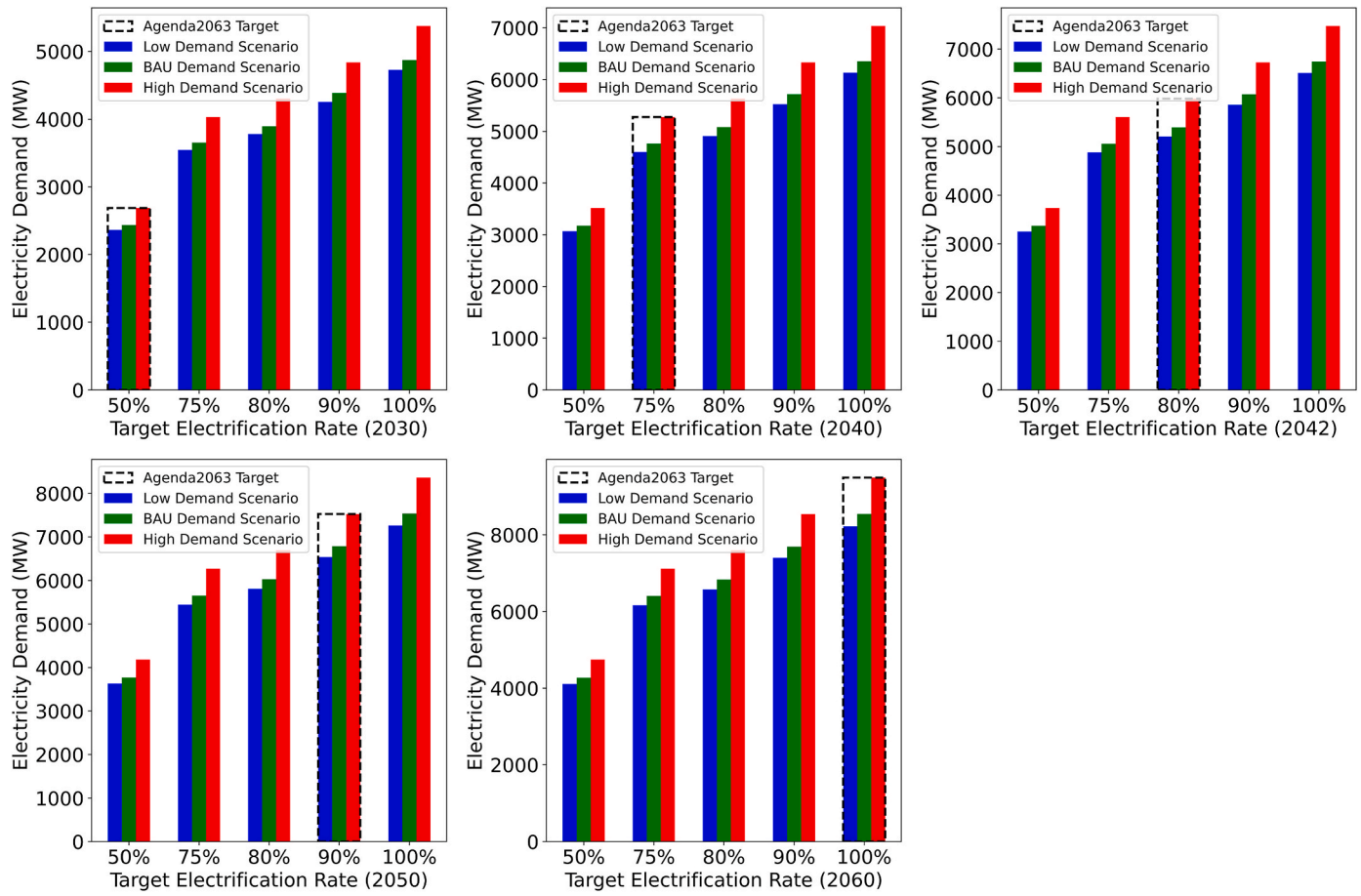


Fig. 8. Projected electricity demand based on specific electrification targets (2030: 50 %, 2040: 75 %, 2042: 80 %, 2050: 90 %, and 2060: 100 %) and compared with other possible electrification scenarios.

while the high demand scenario estimates a peak demand of 2687.01 MW. Additionally, with a 100 % electrification rate by 2030, demand is projected to reach 4870.55 MW under the BAU scenario, with a low demand scenario of 4726.34 MW and a high demand scenario of 5374.03 MW.

- **2040 Forecasts:** By 2040, as electrification reaches 75 %, the BAU scenario forecasts a demand of 4764.02 MW. The demand could vary from 4601.14 MW in the low demand scenario to 5275.91 MW in the high demand scenario. If 100 % electrification is achieved by 2040, demand is projected to reach 6352.02 MW under the BAU scenario, with a low demand scenario of 6134.85 MW and a high demand scenario of 7034.54 MW.
- **2042 Estimates:** In 2042, with an 80 % electrification rate, the BAU scenario estimates the demand at 5395.73 MW. The projected range extends from 5206.52 MW in the low demand scenario to 5979.70 MW in the high demand scenario. If full electrification (100 %) is achieved by 2042, the projected demand reaches 6744.66 MW in the BAU scenario, with a low demand scenario estimate of 6508.15 MW and a high demand scenario estimate of 7474.63 MW.
- **2050 Outlook:** For 2050, aiming for a 90 % electrification rate, the BAU scenario predicts a demand of 6782.66 MW. The potential demand ranges from 6534.71 MW in the low demand scenario to 7525.74 MW in the high demand scenario. In the case of achieving full electrification (100 %) by 2050, demand is projected to reach 7536.29 MW in the BAU scenario, with demand estimates extending from 7260.79 MW in the low demand scenario to 8361.93 MW in the high demand scenario.
- **2060 Projections:** By 2060, with a goal of 100 % electrification, the BAU scenario projects a demand of 8544.64 MW. In this year, the

demand could be as low as 8219.47 MW according to the low demand scenario or as high as 9492.14 MW in the high demand scenario.

3.6. Comparative electricity demand outlook at targeted and full access rates

Malawi's pursuit of universal electricity access by 2060 demands a comprehensive transformation of its energy infrastructure. With electricity demand at just 434.4 MW in 2022 - serving merely 12.4 % of the population - the scale of required investment becomes immediately apparent. The projections reveal striking disparities between targeted and universal access scenarios. Under a Business-as-Usual (BAU) scenario reaching 50 % access by 2030, demand would grow to 2435.27 MW. However, achieving full access (100 %) in the same timeframe would more than double this requirement to 4870.55 MW - a remarkable 1020 % increase from 2022 levels. This dramatic difference underscores how access targets fundamentally shape infrastructure planning.

The analysis shows consistent patterns across subsequent decades. In 2040, the gap between 75 % access (4764.02 MW) and universal coverage (6352.02 MW) demonstrates how population growth, industrialization, and improved living standards compound energy needs. This trend continues in 2042, where demand rises from 5395.73 MW at 80 % access to 6744.66 MW at full electrification. Perhaps most revealing is the 2050 projection: while 90 % access requires 6782.66 MW, the final 10 % of connections demand an additional 753.63 MW, highlighting the disproportionate challenges of last-mile electrification. By 2060, when Malawi aims for complete coverage, demand is projected to reach 8544.64 MW - nearly twenty times the 2022 baseline. These

projections not only quantify the infrastructure challenge but emphasize the critical need for strategic, milestone-driven energy planning. These figures, summarized in Table 2, emphasize the urgent need for long-term planning, diversified energy sources, and international support to meet Malawi's universal access goals in a sustainable manner.

3.7. Demand trends and planning

The analysis predicts a linear increase in Malawi's electricity demand for targeted milestones from 2030 to 2060 as electrification rates progressively rise from 50 % to 100 %. As depicted in Fig. 9, beginning with a 50 % electrification rate in 2030, Malawi would need to increase its power supply by approximately 201.7 MW annually under the BAU Scenario to achieve full electrification by 2060. In contrast, under the low demand scenario, the annual increase would be slightly lower at 193.31 MW, reflecting more conservative growth assumptions, while the high demand scenario anticipates a more aggressive increase of 224.69 MW per year, highlighting potential upper demand limits driven by rapid population growth and industrial expansion. This substantial rise in electricity demand is primarily driven by progressive enhancements in electrification targets set for subsequent years, combined with rapid population growth and economic development. Achieving these ambitious targets poses significant challenges, including the need for substantial capital investment and the expansion of infrastructure capable of supporting increased power production and distribution. Moreover, the development of sustainable energy sources is crucial to meet the growing demand without worsening environmental impacts. The diverse scenarios underscore the importance of strategic planning in the energy sector to balance growth, sustainability, and infrastructure development effectively.

4. Discussions

The findings of this study highlight a significant upward trajectory in electricity demand in Malawi, aligning with the ambitious targets set in Agenda 2063, which aims for complete electrification by 2060, up from a mere 12.4 % in 2020. Notably, the projected demand is expected to surge to approximately 8544.64 MW by 2060, underscoring the need for comprehensive development plans and robust enhancements to the existing infrastructure. Furthermore, under BAU scenario, there is an anticipated annual increase of 201.7 MW in electricity demand, signaling steady growth. In contrast, the low demand scenario forecasts a more modest increase of 193.31 MW per year, while the high demand scenario predicts the most aggressive growth at 224.69 MW per year. These projections emphasize the critical need to scale up energy supply capabilities to meet future demands efficiently.

Malawi's dependency on hydropower, which is susceptible to seasonal and climatic variabilities, highlights the urgency for diversifying energy sources and strengthening grid reliability. This entails integrating renewable resources like solar and wind, though challenges such as high initial investment costs, logistical issues, and persistent corruption could impede progress. Moreover, enhancing the grid capacity will

involve extensive updates to transmission and distribution networks and the integration of smart grid technologies, requiring significant capital investment and strategic planning.

A notable recent advancement is Malawi's grid interconnection with Mozambique, anticipated to substantially augment Malawi's electricity supply by providing broader market access. This interconnection not only boosts electricity imports to meet burgeoning demands but also enhances the stability of Malawi's power system amidst local fluctuations. Additionally, the recent commissioning of large-scale solar PV projects connected to the national grid marks a significant stride towards augmenting RE capacity, further diversifying the energy mix and reducing dependence on hydropower. Critical to the success of these initiatives are partnerships with international donors, private investors, and regional allies, providing essential financial and technical support. Strengthening policy frameworks to attract investments, manage environmental impacts, and align the expansion of the electricity sector with sustainable development goals is imperative.

As highlighted by Chisale et al. [28], corruption and economic challenges significantly affect the development of RE in Malawi, a country also grappling with high poverty levels. Being landlocked, Malawi encounters unique strategic challenges that influence its energy infrastructure and economic opportunities, complicating the attraction of foreign direct investment (FDI) and management of external debt. These factors are critical for accurately forecasting electricity demand, yet our initial analysis did not fully consider them. Key indicators such as the external debt to GDP ratio, FDI, and geopolitical risks play substantial roles in shaping Malawi's energy future.

The external debt to GDP ratio is vital for assessing Malawi's financial health and its ability to fund energy projects [76]. FDI is crucial for capital injections into the energy sector but also introduces dependencies and vulnerabilities, exacerbated by Malawi's limited access to international markets due to its landlocked position [77]. Geopolitical risks could disrupt current and future energy markets, significantly altering investment landscapes and economic forecasts [78]. Future research must integrate these factors to provide a detailed and nuanced understanding of Malawi's electricity demand dynamics. Such integration is essential for robust and resilient energy planning, considering the country's specific challenges.

Furthermore, future electricity demand could deviate significantly from current predictions due to various uncertainties. Geopolitical risks, economic instability, and global energy market fluctuations could unpredictably alter demand trajectories [79,80]. Climate change impacts, including extreme weather and variability in renewable resources, may cause fluctuations in energy availability and consumption [81]. Technological advancements could either drive demand higher or reduce it if adoption is slower than anticipated [82]. Policy changes may accelerate or suppress growth, while misestimations in population growth or urbanization could lead to unexpected demand shifts. Delays in grid infrastructure expansion might constrain growth, while shifts in consumer behavior, such as increased energy efficiency, could reduce demand [82,83]. These uncertainties underscore the importance of adopting flexible and adaptive forecasting approaches to account for potential deviations. Additionally, the forecasting method employed is heavily reliant on the quality and availability of historical data, particularly the absence of long-term hourly load data, which limits temporal granularity and model refinement. These uncertainties and limitations emphasize the need for robust, flexible methodologies that can adapt to data constraints and evolving real-world conditions.

5. Conclusions

This comprehensive study thoroughly forecasts Malawi's electricity demand trajectory aligned with Agenda 2063's ambitious goal of achieving 100 % electrification by 2060. It introduces the ECEEMDAN-BiGRU model, a sophisticated forecasting tool that integrates ECEEMDAN and BiGRU models. This advanced model capably handles the

Table 2

Projected electricity demand (MW) at 100 % electrification.

Target Year	Access Rate (%)	BAU Demand (MW)	Low Demand (MW)	High Demand (MW)
2030	50	2435.27	2363.17	2687.01
	100	4870.55	4726.34	5374.03
2040	75	4764.02	4601.14	5275.91
	100	6352.02	6134.85	7034.54
2042	80	5395.73	5206.52	5979.70
	100	6744.66	6508.15	7474.63
2050	90	6782.66	6534.71	7525.74
	100	7536.29	7260.79	8361.93
2060	100	8544.64	8219.47	9492.14

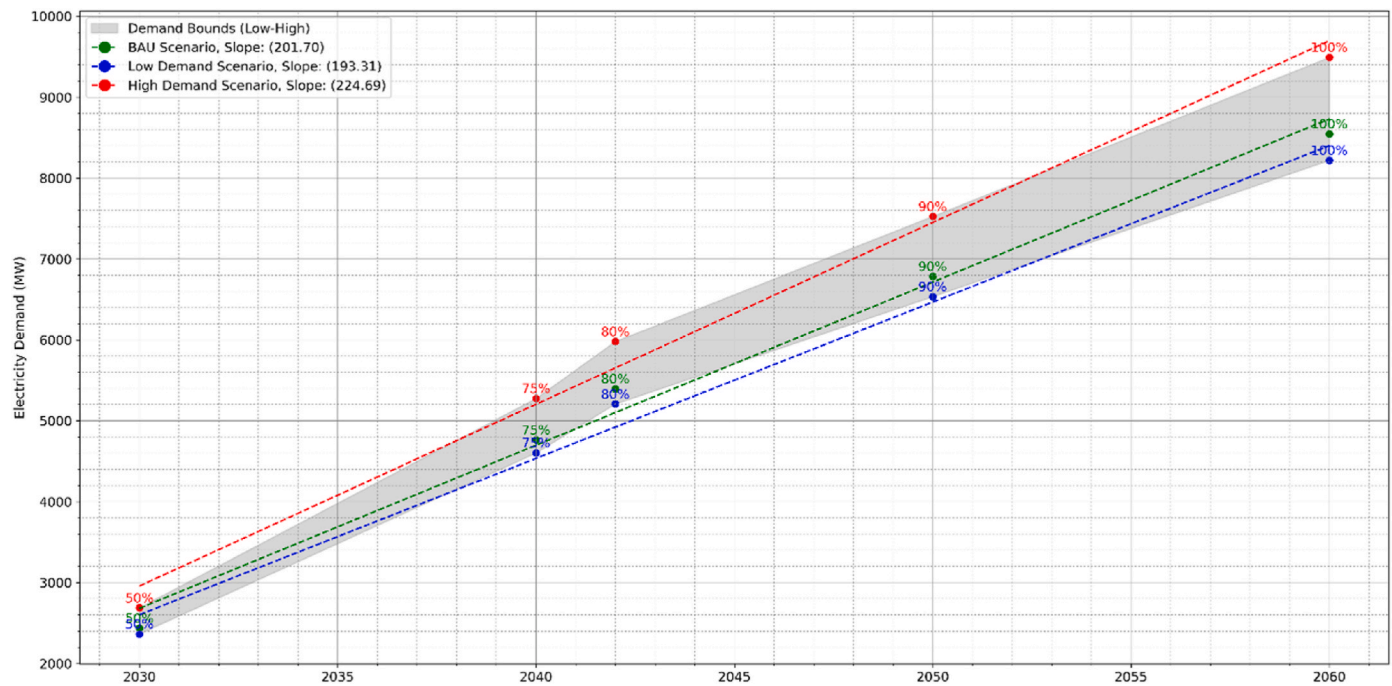


Fig. 9. Electricity demand from 2030 to 2060 with increasing electrification rates across scenarios (BAU, low, and high demand) to determine the required rate of development.

complex dynamics of electricity demand by analyzing temporal patterns and nonlinear relationships inherent in the data. To determine the optimal model configuration, the study evaluated different models for forecasting electricity demand in Malawi, considering climatic and demographic variables. The ECEEMDAN-BiGRU model stands out, achieving the lowest MSE of 52.45 and the highest R^2 of 0.9715, signifying exceptional predictive accuracy. Furthermore, it recorded a MAPE of 0.0205, underscoring its precision in demand forecasting.

The study outlines three primary electricity demand scenarios to anticipate future energy needs in Malawi, each reflecting different growth trajectories influenced by policy and economic factors. The *BAU scenario* serves as the baseline, assuming continued growth based on current trends and existing policy frameworks. The *low demand scenario* projects a more conservative growth path, factoring in potential economic and demographic slowdowns that may dampen energy requirements. Conversely, the *high demand scenario* considers more aggressive growth, fueled by rapid industrialization and population expansion, predicting higher energy demands. These scenarios help in strategizing for future energy policies and infrastructure development.

Targeted projections according to Malawi's agenda 2063 for key years illustrate the potential shifts in demand corresponding to escalating electrification rates.

- **2030 Projections:** Anticipates a 50 % electrification rate with demands varying from 2435.27 MW in the low demand scenario to 2819.14 MW in the high demand scenario, with the BAU Scenario predicting 2687.01 MW.
- **2040 Forecasts:** With a 75 % electrification rate, demand estimates stretch from 4601.14 MW to 5275.91 MW, with the BAU Scenario forecasting 4764.02 MW.
- **2042 Estimates:** With an 80 % electrification goal, demand estimates range from 5206.52 MW in the low demand scenario to 5979.70 MW in the high demand scenario, with the BAU Scenario estimating 5395.73 MW.
- **2050 Outlook:** Targeting a 90 % electrification rate, projected demands vary from 6534.71 MW in the low demand scenario to 7525.74 MW in the high demand scenario, with the BAU Scenario predicting 6782.66 MW.

7525.74 MW in the high demand scenario, with the BAU Scenario predicting 6782.66 MW.

- **2060 Projections:** Aiming for full electrification, demand could range from 8219.47 MW in the low demand scenario to 9492.14 MW in the high demand scenario, with the BAU Scenario projecting 8544.64 MW.

Furthermore, the study highlights that Malawi's electricity demand increasing linearly from 2030 to 2060 as electrification rates rise from 50 % to 100 %. Under the BAU scenario, an annual increase of 201.7 MW is needed to achieve full electrification by 2060, with the low and high demand scenarios requiring 193.31 MW and 224.69 MW per year, respectively. These varied projections emphasize the need for Malawi to scale up its energy infrastructure and diversify its energy sources to meet the anticipated rise in demand. This comprehensive analysis helps policymakers and stakeholders visualize the potential challenges and opportunities as Malawi strives towards complete electrification, supporting strategic planning and investment in the energy sector.

Future research will focus on evaluating optimal energy mixes to meet the projected demand, considering key metrics such as energy production, Levelized Cost of Electricity (LCOE), and CO₂ emissions. Integrating long-term hourly data will enhance the accuracy of demand projections, while sector-specific energy consumption breakdowns will further refine the estimates. Additionally, future work will explore off-grid systems, policy frameworks, climate resilience strategies, and the financing mechanisms required to scale up energy infrastructure. Time series-specific cross-validation techniques, including K-fold variants such as Forward Chaining and Blocked Cross-Validation, will also be investigated to improve model validation and ensure the robustness of long-term forecasting results.

CRediT authorship contribution statement

Sylvester William Chisale: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Han Soo Lee:** Writing – review & editing,

Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Conceptualization. **Manuel Alejandro Soto Calvo:** Writing – review & editing, Visualization, Validation, Software, Methodology, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.energy.2025.137212>.

Data availability

Data will be made available on request.

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